

4-3-1: Examining Digital Certificates

At the end of this episode, I will be able to:

1. Identify and explain the importance of digital certificates, given a scenario.

Learner Objective: *Identify and explain the importance of digital certificates, given a scenario*

Description: In this episode, the learner will examine digital certificates, digital signatures and certificate formats. We will explore the file extensions and where these certificates are utilized.

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- What are digital certificates?
 - An electronic document issued by a trusted authority called a certificate authority (CA)
 - This document binds a public key to an individual, device, or service.
 - These certificates serve as a means of authentication and encryption
 - Digital certificates are used to:
 - Secure communications (HTTPS, SSH, SFTP..)
 - Create digital signatures
 - Access control
 - Can provide strong authentication
 - You mentioned digital signatures; what are they?
 - Digital signature - the use of a private key to create a unique value attached to a document or message
 - This allows anyone with the corresponding public key to verify the signature, ensuring it hasn't been tampered with and was indeed signed by the private key holder
 - What are some types of certificates?
 - *X.509 Certificates:*
 - Usage: Authentication, secure email, SSL/TLS encryption, code signing, VPNs, and more.
 - File Extension: .cer, .crt, .der, .pem, .pfx, .p12 for example
 - *Public-Key Cryptography Standards #7 (PKCS#7) certificates:*
 - Usage: Secure email attachments, code signing, and digital signatures for documents.
 - File Extension: .p7b, .p7c, .p7m
 - *Public-Key Cryptography Standards #12 certificates:*
 - Usage: Typically used for storing private keys and associated X.509 certificates, often for secure authentication.
 - File Extension: .pfx, .p12
 - *Privacy Enhanced Mail (PEM) certificates:*
 - Usage: Often used for SSL/TLS certificates, it can include certificates, private keys, and CA bundles.
 - File Extension: .pem
 - *Distinguished Encoding Rules (DER) certificates:*
 - Usage: Commonly used in binary format for SSL/TLS certificates and code signing.
 - File Extension: .der, .cers
 - *Pretty Good Privacy (PGP) certificates:*
 - Usage: Used for secure email communication and encryption of files and messages.
 - File Extension: .pgp, .asc
 - *Secure/Multipurpose Internet Mail Extensions (S/MIME) certificates:*
 - Usage: Secure email communication and digital signatures for message integrity.
 - File Extension: .p7s
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- Additional Resources:
 - If applicable